

$$3+0+2 = \textcircled{5} + 28 = 33$$

till now so far

CMI
BVR

Probability
Quiz 2

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If you are using a theorem, quote it precisely. Simplify your answer.

[3+4+3]

1. If P is the matrix $\begin{pmatrix} 0 & 1/3 & 2/3 & 0 \\ 2/3 & 0 & 1/3 & 0 \\ 1/3 & 2/3 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ Then

$$\lim_n P^n = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \rightarrow \text{rewrite: } \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Calculation/Reason:

↳ We know that for a connected and aperiodic Markov chain, the limiting matrix is of the form $\begin{pmatrix} \pi \\ \pi \\ \pi \\ \pi \end{pmatrix}$, where π is a probability vector.
or π is a left eigen vector of the matrix of the connected component.

$$P = \begin{pmatrix} 0 & \frac{1}{3} & \frac{2}{3} & 0 \\ \frac{2}{3} & 0 & \frac{1}{3} & 0 \\ \frac{1}{3} & \frac{2}{3} & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$A = \begin{pmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{pmatrix}$ $B = [1]$

The state space has two disjoint connected components, and we can apply the above fact to both of them individually.
(or, in $P = \begin{bmatrix} A & \\ & B \end{bmatrix}$, $P^2 = \begin{bmatrix} A^2 & \\ & B^2 \end{bmatrix}$ $\Rightarrow$ Block matrix multiplication.
 $\rightarrow P^n = \begin{bmatrix} A^n & \\ & B^n \end{bmatrix}$)

↳ $A \rightarrow A$ is a stochastic matrix

↳ observe that $\begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 0 & \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & 0 & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} & 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} = \pi$
 $\rightarrow$ and we know theorem that π is unique probability eigen vector.

$$\lim_{n \rightarrow \infty} A^n = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}, B^n = [1] \text{ clearly, } \therefore \lim_{n \rightarrow \infty} P^n = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

2. X, Y, Z have joint density $f(x, y, z) = 2(x + y + z)/3$ for $0 < x, y, z < 1$.
then the conditional density of Y given $X = 1/2$ is

$$g(y) = \frac{1+2y}{9} \cdot \frac{9}{2} = \frac{1+2y}{2}$$

$$\text{for } 0 \leq y \leq 1$$

Calculation/Reason:

$\hookrightarrow g(y)$ is such a function so that if $y \in (a, b)$

You missed the
main point of conditional
density
Please Think.
What is
 $P(X = \frac{1}{2})$?

Then $P(Y \in (a, b) | X = \frac{1}{2}) = \int_a^b g(y) dy$

$$\frac{P(Y \in (a, b) | X = \frac{1}{2})}{P(X = \frac{1}{2})} = \frac{\int_{y=0}^b \left(\int_{z=0}^1 \frac{2(\frac{1}{2} + y + z)}{3} dz \right) dy}{\int_{y=0}^1 \left(\int_{z=0}^1 \frac{2(\frac{1}{2} + y + z)}{3} dz \right) dy} = \frac{\int_a^b \frac{1+2y}{9} dy}{\int_0^1 \frac{1+2y}{9} dy} = \frac{\int_a^b \frac{1+2y}{9} dy}{\left(\frac{2}{9}\right)}$$

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$\rightarrow$ So clearly, the factor $g = \frac{1+2y}{2}$ needs

$$\text{so } P(Y \in (a, b) | X = \frac{1}{2}) = \frac{\int_a^b \frac{1+2y}{9} dy}{\left(\frac{2}{9}\right)} = \int_a^b \frac{\frac{1+2y}{2}}{\frac{2}{9}} dy = \int_a^b g(y) dy$$

is shown.

3. (X, Y) bivariate normal with $E(X) = 0 = E(Y)$ and $E(X^2) = 1 = E(Y^2)$ and $E(XY) = \rho$. Then the joint density of (X, Y) is given by (simplify, make answer free of matrices).

$$f(x, y) = \frac{1}{\sqrt{2\pi} \sqrt{1-\rho^2}} e^{-\frac{[x, y] \Sigma^{-1} [x, y]^T}{1-\rho^2}} = \frac{1}{\sqrt{2\pi} \sqrt{1-\rho^2}} e^{-\frac{(x^2 + y^2 - 2\rho xy)}{1-\rho^2}}$$

Calculation/Reason: Are you sure?

→ $E(X) = 0 = E(Y)$ and $E(X^2) = 1 = E(Y^2)$

↳ Mean vector

→ Want ~~matrix~~ covariance

$$\Sigma = \begin{bmatrix} E(X^2) & E(XY) \\ E(YX) & E(Y^2) \end{bmatrix} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$$

Is it?
 Yes.

↳ $f(x, y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} [x, y] \Sigma^{-1} [x, y]^T}$

$-x \Sigma^{-1} x^T$

check $(x - \mu)$

But $\mu = \begin{bmatrix} E(X) \\ E(Y) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

→ $|\Sigma| = (1 - \rho^2)$
 $\Sigma^{-1} = \frac{1}{1-\rho^2} \begin{bmatrix} 1 & -\rho \\ \rho & 1 \end{bmatrix}$

$$\frac{[x, y] \begin{bmatrix} 1 & -\rho \\ \rho & 1 \end{bmatrix} [x, y]^T}{1-\rho^2}$$

$$= \frac{[x - \rho y, y - \rho x] \begin{bmatrix} 1 \\ 1 \end{bmatrix}}{1-\rho^2} = \frac{x^2 - \rho xy + y^2 - \rho xy}{1-\rho^2} = \frac{x^2 + y^2 - 2\rho xy}{1-\rho^2}$$

~~Ca b p~~

You should practice writing.